

Maheshwer Reddy Gillela

Data Engineer



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Highly skilled **Data Engineer** and **Business Intelligence Analyst** with **5+ years** of experience in designing, building, and optimizing data pipelines, databases, and ETL processes. Proficient in leveraging cloud platforms (AWS, Azure, GCP), SQL, Python, and big data technologies to deliver scalable and reliable data solutions. Expertise in **data visualization** (Qlik Sense, Power BI), **cross-functional collaboration**, and **process improvement**. Adept at simplifying complex data into actionable insights and driving data-driven decision-making across organizations>

PROFILE SUMMARY:

- **5+ years'** experience in implementing **Azure data solutions**, provisioning storage account, **Azure Data Factory, SQL server, SQL Databases, SQL Data ware-house, Azure Data Bricks and Azure Cosmos DB.**
- Experienced with **Dimensional modeling, Data migration, Data cleansing, Data profiling**, and **ETL Processes** features for data warehouses.
- Expertise in building batch and real-time data processing systems leveraging) **AWS services (S3, Redshift, EMR, Lambda, DynamoDB and Azure services (ADLS, ADF, Databricks, Synapse Analytics).**
- Software development involving cloud computing platforms like **Amazon Web Services (AWS), Azure and Google Cloud (GCP).**
- Collaborated with data scientists and business leaders to **align AI/ML models** with **business goals**, ensuring high-impact data-driven decision-making.
- Deployed **Dockers Engines** in Virtualized Platforms for containerization of multiple apps.
- Developed and deployed **machine learning models** using **TensorFlow, PyTorch, and Spark MLlib** to enhance predictive analytics solutions.
- Extensive work experience in all phases of Software Development Life Cycle (**SDLC**) including Requirement **Analysis**, Design, Coding, Testing and Implementation in **Agile (Scrum), TDD Environment.**
- Proficient in building **CI/CD** pipelines in **Jenkins** using pipeline syntax and **groovy** libraries.
- Excellent knowledge on **Hadoop** Architecture and ecosystems such as **HDFS, Hive, Pig, Sqoop, Job Tracker, Task Tracker, Name Node, Data**
- Adept at configuring and installing Hadoop/Spark Ecosystem Components and Proficient with **Spark Core, Spark SQL, Spark MLlib, Spark GraphX and Spark Streaming** for processing and transforming complex data using in-memory computing capabilities written in **Scala**. Worked with Spark to improve efficiency of existing algorithms using **Spark Context, Spark SQL, Spark MLlib, Data Frame, Pair RDD's and Spark YARN.**
- Good understanding of **Spark** Architecture with **Databricks, Structured Streaming.** Setting Up **AWS** and **Microsoft Azure** with Databricks, **Databricks** Workspace for Business **Analytics**, Manage Clusters in Databricks, Managing the Machine Learning Lifecycle
- Built and consumed Restful **APIs** using **Python** to integrate third-party data sources into the existing data infrastructure.
- Involved in using **Jenkins, Selenium, Docker and Kubernetes** clusters and **power Shell scripting** for efficient data management and decentralized access. Experience working with Front end technologies like **Html, CSS, JS, ReactJS.**
- Integrated on-premise data sources with **GCP** using **Pub/Sub** and **Dataflow.**
- Developed web-based applications using **Python, DJANGO, QT, C++, XML, CSS3, HTML5, DHTML, JavaScript and jQuery.**
- Hands-on experience with **Amazon EC2, Amazon S3, Amazon RDS, VPC, IAM, Amazon Elastic Load Balancing, Auto Scaling, CloudWatch, SNS, SES, SQS, AWS Lambda, EMR** and other services of the **AWS** family.
- Experience in application of various data sources like **Oracle SE2, SQL Server, Flat Files** and Unstructured files into a data warehouse.

- Proficient with **Spark Core, Spark SQL, Spark MLlib, Spark GraphX and Spark Streaming** for processing and transforming complex data using in-memory computing capabilities written in **Scala**. Worked with **Spark** to improve efficiency of existing algorithms using **Spark Context, Spark SQL, Spark MLlib, Data Frame, Pair RDD's and Spark YARN**.
- Hands-on experience with Hadoop architecture and various components such as **Hadoop** File System **HDFS**, Job Tracker, Task Tracker, Name Node, Data Node and **Hadoop MapReduce** programming.

EDUCATION:

- Masters in Computer Science from Bradley University, USA

TECHNICAL SKILLS:

Skill Area	Key Skills
Data Storage	Data Lakes, Data Warehousing, SQL & NoSQL Databases, Cloud Storage
Data Processing	ETL/ELT Development, Data Pipelines, Batch & Stream Processing, Data Transformation
Big Data Technologies	Hadoop, Spark, Distributed Computing, Large-scale Data Processing
Cloud Platforms	Azure, AWS, Google Cloud Platform (GCP)
Programming Languages	Python, SQL, Java, Scala
ETL/ELT Tools	Apache NiFi, Informatica, Talend, Apache Airflow, Azure Data Factory, AWS Glue
Analytics & Reporting	Data Visualization, BI Tools (Tableau, Power BI, QlikView, Qlik Sense), Descriptive & Predictive Analytics , Excel(Pivot tables, VLOOKUP, macros, data analysis
Orchestration	Workflow Automation, Job Scheduling, Data Pipeline Orchestration
Machine Learning	Model Deployment, Feature Engineering, Integration with Data Pipelines
DevOps & CI/CD	Version Control (GitHub, Git , GitHub Actions), Continuous Integration/Continuous Deployment (CI/CD), Infrastructure as Code (IaC)
Networking	Data Integration, API Management, Cloud Networking, Data Migration
Data Quality	Data Profiling, Data Cleansing, Data Validation, Data Lineage
MLOps	MLflow, Kubeflow, TensorFlow Extended (TFX), Model Versioning & Monitoring
Collaboration Tools and Documentation Tools	JIRA, Confluence, Slack, Microsoft Teams , Confluence, SharePoint, Markdown
Data Warehousing	OLAP, Dimensional Modeling, Schema Design, Data Marts, Snowflake, T-SQL, DBT, Fivetran

WORK EXPERIRNCE:

Client: Northern Trust Corporation, Chicago, Illinois, USA

Apr 2023 - Present

Role: Azure Data Engineer

Description: Northern Trust Corporation is a leading financial services specializing in wealth management, asset servicing, asset management, and banking solutions. Developed & maintained scalable data architectures and pipelines to support analytics and reporting needs, collaborating with stakeholders to ensure compliance, security, & performance across the organization.

Responsibilities:

- Design, develop, and maintain scalable data architectures using **Azure Data Lake**, **Azure SQL Database**, and **Azure Synapse Analytics**.
- Collaborated with **business analysts** and **senior leadership** to design and deliver **data-driven solutions** that aligned with organizational goals, improving operational efficiency by **20%**.
- Developed and optimized **ELT** pipelines in **Snowflake** using **DBT** and **Fivetran**, improving data ingestion reliability and processing speed. Used **T-SQL** for writing complex stored procedures, views, and queries in **Azure SQL Database** to support data transformation and reporting needs.
- Tuned **Snowflake** queries using clustering keys, partitions, and caching to enhance performance across large datasets. Maintained secure access control in **Snowflake** using role-based access and implemented best practices for **data governance**.
- Automate data transformation, cleansing, and loading into **Azure storage and databases** for use in analytics and reporting.
- Implement and optimize data storage solutions using **Azure Data Lake Storage**, **Azure Blob Storage**, and **Azure SQL Database**.
- Use **Azure Databricks** and **Apache Spark** to process and analyze large datasets for real-time insights and analytics.
- Design and **deploy APIs** for data integration using **Azure API Management** and **Azure Logic Apps** to integrate data across systems and applications.
- Monitor data pipelines and infrastructure performance using **Azure Monitor**, **Log Analytics**, and **Azure Application Insights**.
- Implement security best practices in data management by using **Azure Active Directory (AAD)**, **Azure Key Vault**, and **Azure Security Center**.
- Implemented model monitoring for drift detection, versioning, and retraining triggers using **Prometheus**, **Grafana**, and **ELK Stack**.
- Ensured compliance with **CCPA** regulations alongside **GDPR** and **HIPAA** by implementing **robust data governance** policies and security measures
- Design and develop **interactive dashboards and reports** using **Power BI** to deliver insights to stakeholders across the organization, such as wealth management, risk analysis, and investment performance.
- Develop and maintain **CI/CD** pipelines for data engineering projects using tools like **Azure DevOps**, **GitHub Actions**, or **Jenkins**. Ensure automated build, test, and deployment processes for data pipelines and infrastructure.
- Develop **Python scripts** for automating data extraction, transformation, and loading (**ETL**) processes, as well as for interacting with **Azure services**.
- Developed and maintained **CI/CD pipelines** using tools like **Azure DevOps** and **GitHub Actions** to automate the deployment and integration of data engineering solutions, ensuring efficient and reliable delivery of **Azure data pipelines**, **Power BI** reports, and infrastructure changes.
- Created interactive dashboards in **Qlik Sense** to visualize key metrics and trends for wealth management and risk analysis.

Environment: **Azure Data Lake**, **Azure SQL Database**, **Azure Synapse Analytics**, **ETL/ELT pipelines**, **Azure Data Factory**, **Azure Blob Storage**, **Azure SQL Database**, **APIs**, **Apache Spark**, **Azure Logic Apps**, **Azure Active Directory (AAD)**, **Azure Key Vault**, **Azure Security Center**, **CI/CD**, **Azure DevOps**, **GitHub Actions**, **GDPR**, **Power BI**, **Azure Monitor**.

Client: **Arthur J. Gallagher & Co**, **Rolling Meadows**, **Illinois**, **USA**

Nov 2021 - Mar 2023

Role: **AWS Data Engineer**

Description: **Arthur J. Gallagher & Co.** is a global insurance brokerage, risk management, and consulting firm specializes in property and casualty insurance, employee benefits, and reinsurance services. Managed the design and implementation of cloud infrastructure to support critical applications, ensuring automation, security, and scalability while optimizing performance and compliance with regulatory standards.

Responsibilities:

- Design, implement, and manage **AWS** cloud infrastructure using services like **EC2**, **VPC**, **S3**, and **RDS** to support business-critical applications.
- Automate the provisioning of infrastructure using **AWS CloudFormation** or **Terraform**, ensuring infrastructure is version-controlled, auditable, and easily reproducible.

- Implement **serverless** architectures using **AWS Lambda** to improve efficiency and reduce costs.
- Develop and maintain **CI/CD pipelines** using tools such as **AWS CodePipeline**, **CodeBuild**, and **CodeDeploy** to automate the deployment process of applications and services.
- Developed and managed real-time data streaming solutions using **Kafka** and **Spark Streaming**, enabling low-latency processing and high-throughput **data pipelines** for analytics and reporting.
- Integrate **Git** for source control and use **Jenkins** or **AWS CodePipeline** for continuous integration and deployment.
- Designed and deployed **Qlik Sense** dashboards to provide real-time insights into insurance claims and risk management data, improving decision-making efficiency by **25%**.
- Ensure compliance with regulatory requirements (such as **PCI DSS** or **GDPR**) by implementing security best practices, such as using **AWS IAM**, **AWS KMS**, and **AWS Config** for identity, access management, encryption, and configuration management.
- Implement monitoring and logging solutions using **Amazon CloudWatch**, **AWS X-Ray**, and **AWS CloudTrail** to monitor the performance, availability, and security of cloud resources.
- Integrated **MLflow** for **model tracking** and automated deployment in **Azure ML** and **AWS SageMaker**.
- Design and manage scalable storage solutions using **Amazon S3**, **EBS**, and **Glacier** for data archiving and backups.
- Implement backup and disaster recovery strategies using **AWS Backup** and **AWS Elastic Disaster Recovery**.
- Integrated **Kubernetes** with cloud-native services, such as **AWS EKS** and **GCP GKE**, to leverage additional scalability and managed services. Developed **Kibana** Dashboards based on the Log stash data and Integrated different source and target systems into **Elasticsearch** for near real time log analysis of monitoring End to End transactions.
- Utilize **Python libraries like Pandas, NumPy, and PySpark** for data manipulation, analysis, and transformation.
- **Designed and implemented a scalable real-time data processing architecture** leveraging **Amazon Kinesis** for ingesting high-velocity streaming data and **Amazon Athena** for querying structured datasets stored in S3. Utilized robust **data modeling** techniques, including partitioning and schema design, to optimize query performance while adhering to **data security** standards like encryption, role-based access control (**RBAC**), and **GDPR** compliance.
- Set up and worked on **Kerberos** authentication principals to establish secure network communication on cluster and testing of **HDFS**, **Hive**, **Pig** and **MapReduce** to access cluster for new users.
- Write complex **SQL** queries to extract, aggregate, and analyze data from relational databases like **Amazon RDS (MySQL, PostgreSQL)** and **Amazon Redshift**

Environment: EC2, VPC, S3, RDS, AWS Cloud, Terraform, AWS Lambda, CI/CD pipelines, AWS CodePipeline, Git, Jenkins, PCI DSS, GDPR, AWS IAM, AWS KMS, AWS X-Ray, EBS, Glacier, Kubernetes, Kibana, Python libraries like Pandas, NumPy, and PySpark, SQL, PL/SQL, Kerberos, HDFS, Hive, Pig, MapReduce, Amazon Redshift.

Client: Medicare, Bangalore, India

Aug 2020 - June 2021

Role: GCP Data Engineer

Description: Medicare is a federal health insurance program provides coverage to eligible individuals. Contributed to the design and management of scalable data infrastructure across cloud platforms, ensuring efficient data processing, security, and compliance with industry regulations.

Responsibilities:

- Design, deploy, and manage data infrastructure on **Google Cloud Platform (GCP)**, including services like **Google BigQuery**, **Google Cloud Storage**, and **Google Cloud SQL**.
- Ensure that the data infrastructure scales effectively with the company's growing data needs and workloads.
- Develop and maintain **ETL** (Extract, Transform, Load) pipelines using tools like **Google Cloud Dataflow**, **Google Cloud Composer**, or **Apache Airflow**.
- Utilized **Redis** and **MongoDB** for caching and **NoSQL data storage**, improving query performance and supporting low-latency applications
- Implement security measures to protect data, including encryption, access controls, and compliance with relevant regulations (e.g., **GDPR**, **HIPAA**)

- Have written **pythonDAGS** in airflow which orchestrate end to end data pipelines for multiple application and involved in setting up of apache airflow service in **GCP**.
- Manage and optimize **Azure cloud services**, including **Azure Data Lake Storage, Azure SQL Database, Azure Synapse Analytics, and Azure Blob Storage**.
- Experience in moving data between **GCP** and **Azure** using **Azure Data Factory**.
- Knowledge about **cloud dataflow** and **Apache beam**. and Good knowledge in using cloud shell for various tasks and deploying services

Environment: Google Cloud Platform, Google BigQuery, Google Cloud Storage, Google Cloud SQL, ETL, Google Cloud Dataflow, Google Cloud Composer, Apache Airflow, GDPR, HIPAA, pythonDAGS, Azure cloud services, Azure Data Lake Storage, Azure SQL Database, Azure Synapse Analytics, Azure Blob Storage, GCP, Azure Data Factory Apache.

Client: Ford Motor Company, Bangalore, India

June 2019 - July 2020

Role: Data Engineer

Description: **Ford Motor Company** is an iconic American multinational automaker known for its extensive range of vehicles and its significant impact on the automotive industry. I involved in managing, processing, and analyzing large volumes of data to support various aspects of the automotive business.

Responsibilities:

- Design and implement scalable data architectures to manage and optimize the flow of data across various systems within the manufacturing environment
- Use tools and technologies such as **Apache Spark, Apache Kafka, or custom ETL** solutions to ensure seamless data integration and processing.
- Use database systems like **SQL Server, PostgreSQL, or cloud-based** solutions like **Azure Redshift or Google BigQuery** to manage and store data.
- Wrote and executed various **MYSQL** database queries from **Python** using **Python-MYSQL connector** and **MySQL dB package**.
- Used **Jira** for ticketing and tracking issues and **Jenkins** for continuous integration and continuous deployment.
- Implemented **Azure data lake, Azure Data factory** and **Azure data bricks** to move and conform the data from on - premises to cloud to serve the analytical needs of the company. Used **Docker** for managing the application environments.
- Skilled in monitoring servers using **Nagios, Cloud watch** and using **ELK Stack- Elastic search** and **Kibana**.
- Implemented a continuous delivery (**CI/CD**) **pipeline** with **Docker** for custom application images in the cloud using **Jenkins**.

Environment: Apache Spark, Apache Kafka, SQL Server, PostgreSQL, Azure Redshift , Google BigQuery, MYSQL, Python, MySQL dB package, Jira, Jenkins , Azure, Docker, Elastic search, Kibana, CI/CD) pipeline.